

Thibaut GERMAIN

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Paris, France

EXPERIENCE

- 03/2025 - today **Postdoctoral researcher, Centre de Mathématiques Appliquées de Polytechnique (CMAP), Palaiseau, France.** Postdoctoral researcher within the European project [ELIAS](#), developing sustainable artificial intelligence to support sustainable development. My research focuses on learning from populations of dynamical systems, leveraging shared structures to improve the understanding, representation, and estimation of individual systems. Grounded in operator learning, this work extends to conditional expectation operators and their applications in statistical learning. Conducted in collaboration with [Vladimir KOSTIC](#), [Karim LOUNICI](#), and [Rémi FLAMARY](#).
- 01/2025 - 02/2025 **Postdoctoral researcher, Centre Borelli, Ecole Normale Supérieure Paris-Saclay, Gif-sur-Yvette, France.** Continuation of my PhD research in a postdoctoral position for two months, focusing on adapting some of the developed methods to deep learning approaches. It had led to one publication.
- 07/2024 **Invited researcher, Duke University, Durham, NC (USA).** One-month research stay. Contributed to the development of tools for analyzing therapeutic progress in eating disorders from video data. Work focusing on detecting and quantifying facial emotion by combining signal processing and shape analysis methods.
- 01/2017 - 07/2019 **Supply Chain Engineer, Tempest Tech Corp., Fresno, CA (USA).** Firefighting equipment manufacturer. Main responsibilities included optimizing operational processes (production and logistics), supervising on-site research and development activities, and contributing to team management, including production team oversight, staff training, and recruitment.

TRAINING

- 10/2021 - 12/2024 **PhD in Applied Mathematics, Centre Borelli, Ecole Normale Supérieure Paris-Saclay, France.** Research focusing on shape detection and shape analysis for physiological time series. Link to the thesis manuscript: <https://theses.fr/s298010>. Co-supervised by [Laurent OUDRE](#) and [Charles TRUONG](#). Thesis defended after 37 months on December 6th, 2024 before a committee including:
- Jury chair:
 - [Alain TROUVE](#) (Professor, ENS Paris-Saclay)
 - Reviewers:
 - [Jessica LIN](#) (Associate professor, George Mason University)
 - [Romain TAVENARD](#) (Professor, Université Rennes 2)
 - Examiners :
 - [Laurent YOUNES](#) (Professor, Johns Hopkins University)
 - [Guillermo SAPIRO](#) (Professor, Princeton University)
 - PhD supervisors:
 - [Laurent OUDRE](#) (Professor, ENS Paris-Saclay)
 - [Charles TRUONG](#) (Assistant professor, Université Américaine de Paris)
- 2021-2020 **Master of Science, Ecole Normale Supérieure Paris-Saclay, France.** Master Mathématiques Vision et Apprentissage (MVA), Highest honors.
- 2017-2016 **Master of Science, Université Lyon 1, France.** Master on operational research and supply chain.
- 2017-2014 **Engineer diploma, Ecole des Mines de Saint-Etienne, France.** Master on logistic and production management.
- 2014-2012 **Classe préparatoire, Lycée Victor Hugo, Besançon, France.** MPSI/MP*

TEACHING

- 2025 - 2026 **Statistic and probability refresher, Ecole Polytechnique, M2 [data science](#), 24h eq. CM.**
Designed and taught a refresher course covering measure theory, random variables, modes of convergence, parametric statistical models, point estimation, optimality, and hypothesis testing.
- 2025 **Programming with python, Ecole CentraleSupélec, M2 [Exed](#), 24h ep. CM.**
Improved and taught a course on Python programming for machine learning. The class was interactive, combining practical and live coding sessions.
- 2021 - 2024 **Introduction to statistic and machine learning, ENS Paris-saclay, M1, 66h ep. TD.**
Led and designed the set of exercises and coding workshops covering statistics (estimation, optimality, testing, linear models) and machine learning (generalization bounds, SVMs).

- 2021 - 2024** **Agrégation de mathématiques, option probability and statistic (option A) – Oraux blancs., ENS Paris-saclay, M2, 70h eq. TD.**
 Led practical sessions on sampling, convergence, estimation, linear models, and hypothesis testing. Served on the mock oral examination jury.
- 2022** **Introduction measure theory and probability, ENS Paris-saclay, niveau L3, 14 eq. TD.**
 Led practical sessions covering random variables, convergence, and conditional expectations.

STUDENTS

PhD Student

- Valerio GUERRINI, 2024 - , co-supervised with [Charles TRUONG](#) and [Laurent OUDRE](#), working on “Unsupervised approaches for constructing semantic summaries from physiological signals. Application to behavior analysis”.

Master Students

- Sami Chemlal, M1 CentraleSupélec, April–September 2026, co-supervised with [Rémi FLAMARY](#) and [Karim LOUNICI](#), on “Multitask and dictionary learning for conditional expectation operator learning”.
- Valerio GUERRINI, M2 MVA, April–September 2024, co-supervised with [Charles TRUONG](#) and [Laurent OUDRE](#), on “A benchmark for discovering recurrent patterns in time series”.
- Lucas HAUBERT, M2 MVA, April–September 2024, co-supervised with [Chrysoula KOSMA](#) and [Laurent OUDRE](#), on “Shape-based loss functions for deep learning on time series”.

Undergraduate Students

- Sienna O’Shea, BSc Math & CS, Ecole Polytechnique, February–June 2026, co-supervised with [Rémi FLAMARY](#), on “Implementation of spectral-OT metrics in the optimal transport library POT”.
- Romain DELAUNAY and Hugo GAIBLE, L3 ENS Paris-Saclay, April–June 2023, co-supervised with [Alexandre BOIS](#) and [Laurent OUDRE](#), on “Convolutional dictionary learning for time series”.
- William RAMOS and [Oussama ZEKRI](#), L3 ENS Paris-Saclay, April–June 2022, co-supervised with [Laurent OUDRE](#), on “Optimal transport and elastic measures for time series”.

RESPONSIBILITIES

Seminar

- 2022 - 2024** **"Panorama Recherche", ENS Paris-saclay.**
 Intended for incoming undergraduate students (L3) in the Mathematics Department. Its goal is to provide a broad overview of research in applied mathematics through presentations given by researchers from various fields. I was responsible for ensuring the smooth running of the seminar, welcoming and introducing invited speakers, and facilitating the discussion session.

Area Chair

2026 ICLR

Reviewer

2026 NeurIPS
2025 ICLR, NeurIPS (best reviewer awards), EJS (Electronic Journal of Statistics)
2024 NeurIPS

SOFTWARE DEVELOPMENT

All of my research projects have led to open-source Python implementations, available on my [GitHub](#). Several of these projects have been packaged into Python libraries or web applications, including:

1. **Python Optimal Transport (POT):** Contributor to a Python library for optimal transport. [\[link\]](#)
2. **Interactive time series motif discovery:** A web application for discovering and visualizing motifs in time series. [\[link\]](#) [\[paper\]](#)
3. **Time series motif discovery library (TSMD):** A Python library for benchmarking time series motif discovery algorithms, including implementations of algorithms, datasets, and evaluation metrics. [\[link\]](#) [\[paper\]](#)
4. **Shape analysis for time series library (TS-PCA):** A Python package for unsupervised representation learning and principal component analysis of irregularly sampled time series with variable lengths, grounded in shape analysis methodology. [\[link\]](#) [\[paper\]](#)
5. **Ventilation behavior analysis tool:** A pilot project in online industrial data science, developed in collaboration with web developers. [\[paper\]](#)

PUBLICATIONS

Preprints:

1. Sami Chemlal, Thibaut Germain, Rémi Flamary, Vladimir R Kostic, and Karim Lounici. "Multi-Task Learning of Conditional Mean Operators: applications to dynamical systems and uncertainty quantification". In: *arXiv preprint* (Sept. 2026). DOI: [10.48550/arXiv.2609.35429](https://doi.org/10.48550/arXiv.2609.35429). URL: <https://arxiv.org/abs/2609.35429>
2. Thibaut Germain, Sami Chemlal, Rémi Flamary, Vladimir R Kostic, and Karim Lounici. "Geometric Dictionary Learning of Dynamical Systems with Optimal Transport". In: *arXiv preprint arXiv:2605.18276* (2026)

Journal papers:

1. Valerio Guerrini, Thibaut Germain, Charles Truong, Laurent Oudre, and Paul Boniol. "Time Series Motif Discovery: A Comprehensive Evaluation". In: *Proceedings of the VLDB Endowment* 18.7 (2025)
2. Konstantin Petrov, Oksana Lenina, Jacqueline Leroy, Véronique Bernard, Thibaut Germain, Charles Truong, Leniz Nurullin, Guzel Sibgatullina, Kinji Ohno, Dmitry Samigullin, et al. "An $\alpha 7$ nicotinic and GABAB receptor-mediated pathway controls acetylcholine release in the tripartite neuromuscular junction". In: *The Journal of Physiology* 603.2 (2025), pp. 507–527
3. Thibaut Germain, Charles Truong, and Laurent Oudre. "Persistence-Based Motif Discovery in Time Series". In: *IEEE Transactions on Knowledge and Data Engineering (TKDE)* 36.11 (2024), pp. 6814–6827. DOI: [10.1109/TKDE.2024.3417303](https://doi.org/10.1109/TKDE.2024.3417303)
4. Thibaut Germain, Charles Truong, Laurent Oudre, and Eric Krejci. "Unsupervised classification of plethysmography signals with advanced visual representations". In: *Frontiers in Physiology* 14 (2023). ISSN: 1664-042X. DOI: [10.3389/fphys.2023.1154328](https://doi.org/10.3389/fphys.2023.1154328)

International conference papers:

1. Valerio Guerrini, Thibaut Germain, Charles Truong, and Laurent Oudre. "Discovering Multiple Subdimensional Motifs in Multivariate Time Series". In: *European Signal Processing Conference (EUSIPCO)*. 2026
2. Thibaut Germain, Rémi Flamary, Vladimir R Kostic, and Karim Lounici. "A Spectral-Grassmann Wasserstein metric for operator representations of dynamical systems". In: *International Conference on Learning Representations (ICLR)*. 2026
3. Thibaut Germain, Chrysoula Kosma, and Laurent Oudre. "Time series representations with hard-coded invariances". In: *Forty-second International Conference on Machine Learning*. 2025
4. Thibaut Germain, Samuel Gruffaz, Charles Truong, Laurent Oudre, and Alain Durmus. "Shapes analysis for time series". In: *Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS)*. 2024, pp. 95607–95638
5. Thibaut Germain, Charles Truong, and Laurent Oudre. "Interactive Motif Discovery in Time Series with Persistent Homology". In: *Machine Learning and Knowledge Discovery in Databases. Research Track and Demo Track (ECML PKDD)*. Cham: Springer Nature Switzerland, 2024, pp. 383–387. ISBN: 978-3-031-70371-3. DOI: [10.1007/978-3-031-70371-3_25](https://doi.org/10.1007/978-3-031-70371-3_25)
6. Thibaut Germain, Charles Truong, and Laurent Oudre. "Linear-trend normalization for multivariate subsequence similarity search". In: *2024 IEEE 40th International Conference on Data Engineering Workshops (ICDEW)*. IEEE. 2024, pp. 167–175. DOI: [10.1109/ICDEW61823.2024.00028](https://doi.org/10.1109/ICDEW61823.2024.00028)
7. Thibaut Germain, Charles Truong, Laurent Oudre, and Eric Krejci. "Unsupervised study of plethysmography signals through DTW clustering". In: *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE. 2022, pp. 3396–3400. DOI: [10.1109/EMBC48229.2022.9870907](https://doi.org/10.1109/EMBC48229.2022.9870907)

National conference papers:

1. Thibaut Germain, Alexandre Bois, Charles Truong, Brian Tervil, and Laurent Oudre. "Détection non supervisée de motifs sur séries temporelles". In: *2023 29ème Colloque Francophone de Traitement du Signal et des Images (GRETSI)*. 2023
2. Thibaut Germain, Charles Truong, Laurent Oudre, and Eric Krejci. "Approches non-supervisées et non-linéaires pour l'analyse de signaux de pléthysmographie". In: *2022 28ème Colloque Francophone de Traitement du Signal et des Images (GRETSI)*. 2022

Thesis:

1. Thibaut Germain. "Pattern detection and shape analysis for physiological timeseries". PhD thesis. Université Paris-Saclay, 2024

TALKS

1. "Dictionary Learning of Dynamical Systems with Optimal Transport" ([New Trends in Statistical Learning 2026](#), Porquerolles, June, 2026)
2. "Dictionary Learning of Dynamical Systems with Optimal Transport" (Machine Learning and Dynamical System Seminar, online, June, 2026; [video](#))
3. "Shape-based embeddings of time series" (L2S Seminar, CentraleSupélec, March, 2026)
4. "Comparing dynamical systems with Optimal Transport" ([CSD seminar](#), ENS Ulm, February, 2026)
5. "Comparing dynamical systems with Optimal Transport" ([SGM seminar](#), École Polytechnique, January, 2026)
6. "Shape and operator embeddings of sequential data for machine learning" (INRIA Rennes, [MALT](#), November 2025)

7. "Spectral-Grassmann Wasserstein metric for operator representations of dynamical systems" (INRIA Saclay, [MIND](#), October 2025)
8. "Invariances and shape embedding for time series machine learning" (EDF R&D, May 2025)
9. "Detection and statistical analysis of temporal patterns" ([R&D Cognition Day](#), January 2025)
10. "Shape analysis and machine learning for time series" ([Duke University](#), July 2024)
11. "Persistence-based motif discovery in time series" ([Centre Borelli](#), June 2024)
12. "Linear-trend normalization for multivariate subsequence similarity search" ([ICDE 2024](#), May 2024)
13. "Interpretable classification of ventilation behaviors based on machine learning" ([Integrative Neuroscience and Cognitive Center \(INCC\)](#), January 2023)
14. "A method for clustering respiratory behaviors based on DTW distances between time series & how new tools enable rediscovery of data" ([Maths & Medicine](#), December 2022)
15. "Unsupervised study of plethysmography signals through DTW clustering" ([EMBC 2022](#), July 2022)